

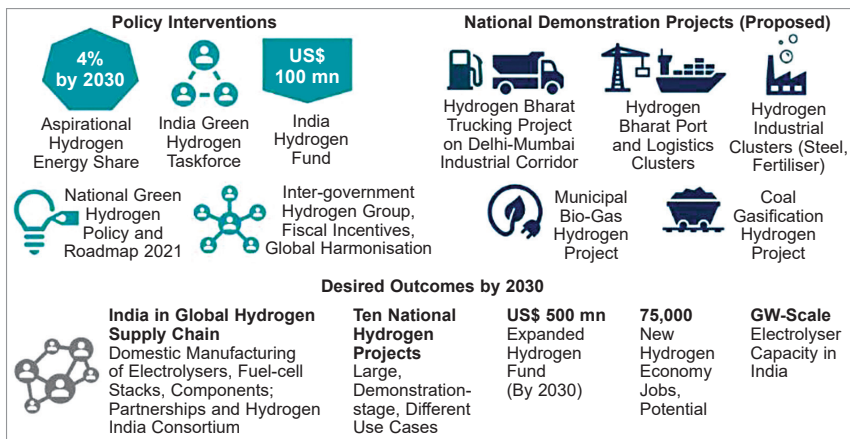


Fig. 8: India's green hydrogen roadmap (Source: <https://www.consultancy.in>)

production can be stored and released when the demand is increasing.

Energy storage and grid balancing.

Energy storage and grid balancing are essential components of a reliable and resilient energy system. GH can act as a valuable energy storage medium, especially in regions with high renewable energy generation. The excess electricity produced can be used for electrolysis of water to produce GH. When the demand for electricity rises or renewable energy production decreases, the stored hydrogen can be converted back into electricity. This approach can enhance the grid stability, reduce the need for fossil fuel-based peaking power plants, and help integrate renewable energy into the grid. GH can also play a pivotal role in addressing the seasonal variation in energy supply, ensuring a consistent and reliable energy system round the year.

Export potential. GH has significant potential as an export commodity.

Countries with abundant renewable energy resources, such as India, Australia, and the Middle East, are well-positioned to become major exporters to countries like Japan and South Korea, which are exploring import of GH for various applications. Exporting GH provides opportunity to foster economic growth, enhance energy security, and strengthen international cooperation.

The Indian initiatives

India has been actively promoting GH as part of its strategy to achieve the climate goals and transit to a cleaner and more sustainable energy system. A comprehensive policy framework is being developed to support the production, storage, transportation, and usage of GH.

This framework includes incentives, subsidies, R&D activities, and regulatory aspects. A roadmap for the development, deployment, and

promotion of GH has been planned. The National Hydrogen Energy Mission (NHEM) (see Box 2) has been initiated to scale up hydrogen production and usage across sectors and facilitate the development of a green hydrogen economy in India.

The government and the industry are investing heavily to improve the efficiency and cost-effectiveness of the production technologies, develop alternative methods for production, produce advanced electrolysers, and address the challenges related to storage and transportation. The Department of Science and Technology and the Department of Biotechnology are funding private and state-owned projects and initiating pilot projects, such as the 'Green Hydrogen Demonstration Project' at Mathura.

Reliance. Reliance Industries Limited (RIL), the largest oil and gas company in India, is investing ₹4000 billion (US\$49 billion) over the next 10 years to create the world's largest GH ecosystem with a production capacity of 1 million tons (907,184 metric tons) per year. It has set up The Dhirubhai Ambani Green Energy Integrated Giga Complex (DAGEIGC) in Gujarat, by investing US\$10 billion to achieve the target of net-zero carbon by 2035 and bring down the cost of GH to 1US\$/kg, compared to the present cost of 8 to 9US\$/kg. The complex is among the largest integrated renewable energy research and development and manufacturing facilities in the world. It will facilitate the production of GH, electrolysers, integrate advanced energy storage systems, fuel cells, and basic and auxiliary materials. RIL is planning to set up 5000 retail outlets over the next three years. It has developed hydrogen-internal combustion engine (H2-ICE) (see Box 3) based on the hydrogen fuel cells technology. The engine is to be used for heavy-duty trucks, buses, and luxury coaches.

Adani New Industries (ANIL). The green hydrogen platform of Adani

Box 2: National hydrogen energy mission

The National Hydrogen Mission announced in 2021 has the overarching objectives of making India the global hub for production, usage and export of GH, and becoming *aatmanirbhar* (self-reliant) in the energy sector. The mission aims to generate ₹8000 billion investment to create production capacity of 5 million metric tonne (MMT) per annum, with capability of increasing it to 10MMT by 2030; generate over 600,000 employment opportunities; and avert 50MMT of CO₂ emissions per year. The initial phase of the mission (2022-23 to 2025-26) will focus on laying the foundation for the R&D facilities; increasing electrolyser manufacturing capacity; accelerating green transition in various industries; and creating a framework for harmonising the regulations and standards with the international norms. The final phase (2026-27 to 2029-30) is going to focus on scaling up the R&D activities; increasing the exports; enhancing decarbonisation across the sectors like railways and aviation that are difficult to decarbonise, promoting pilot projects; and setting up supply and refuelling infrastructure for GH.